



QUANTIFYING THE IMPACT OF FREEDOM ON FINANCIAL INCLUSION

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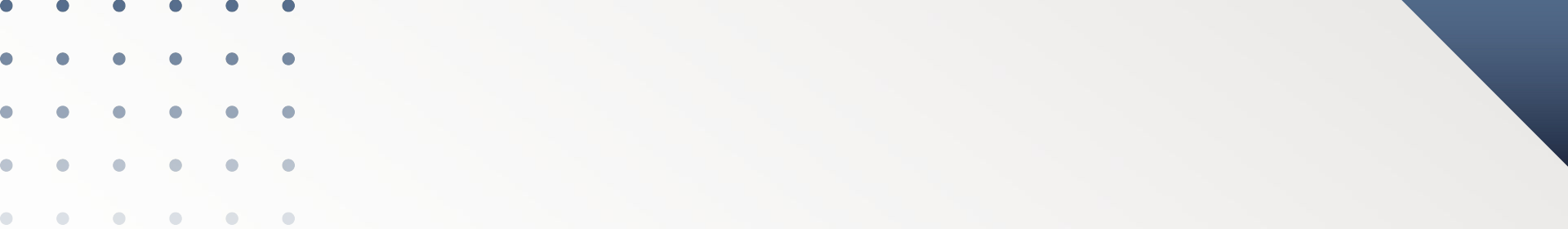
Overarching Theme

How can we draw parallels between the relationship of ***the people and their government*** to the relationship of ***people and their financial institutions?***



Key Questions





QUALITATIVE CONCEPTUALIZATION

What is Financial Inclusion?



Definition: How accessible are financial products and services accessible to individuals and businesses?

Ex. How many people have a bank account? Do women have access to bank loans? Are financial literacy initiatives effective? Do citizens have credit cards?

Why Should We Care?

IMF International Monetary Fund

Central Bank Digital Currencies Can Boost Middle East's Financial Inclusion, Payment Efficiency.

19 countries in the Middle East and Central Asia are exploring issuing a CBDC.

3 weeks ago



United Nations Development Programme

UNDP Partners with the DFINITY Foundation to Enhance Financial Inclusion of MSMEs

ZURICH, SWITZERLAND, JULY 3rd, 2024 – UNDP is pleased to announce a new partnership with the DFINITY Foundation, which will leverage the...

4 days ago



U.S. Agency for International Development (.gov)

USAID Launches Rural Financial Inclusion Program in Guatemala | Latin America & the Caribbean | Press Release

USAID Administrator Samantha Power announced today that, working with Congress, USAID will provide an initial investment of over \$6 million...



CSIS | Center for Strategic and International Studies

The Role of Fast Payment Systems in Addressing Financial Inclusion

Please join the CSIS Scholl Chair in International Business for the report launch of its latest paper Uptake, Use, and Inclusion Gains from...

Apr 19, 2024



CARE - Fighting Global Poverty and World Hunger

CARE invests in Ugandan Fintech Ensibuuko to boost Financial Inclusion for Women - CARE

CARE is pleased to announce a strategic partnership and investment in Ensibuuko, a Uganda-based fintech company.

2 weeks ago



Goldman Sachs Asset Management

Investing in Financial Inclusion: Driving Growth and Building Wealth

Financial inclusion is a critical component of inclusive growth. Growing awareness of its importance is spurring the development of tech-based...

Apr 10, 2024



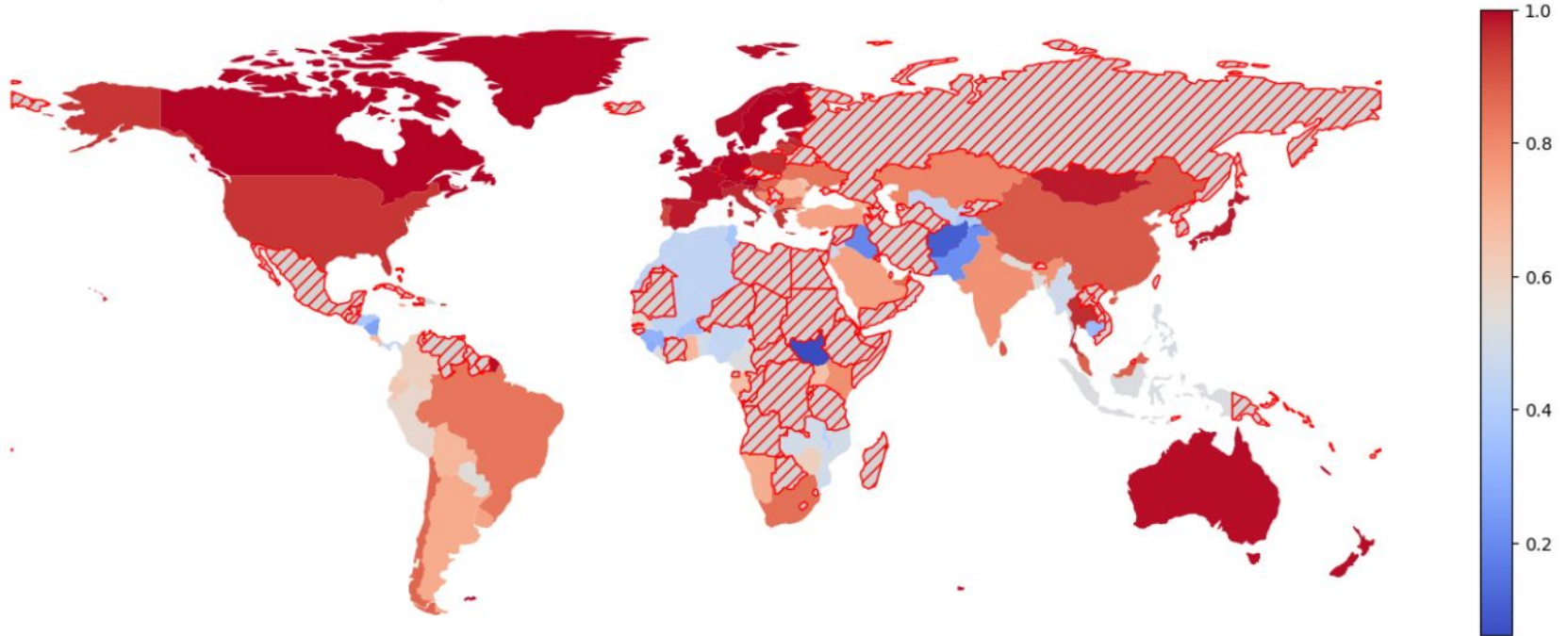
Data Sets

Global Findex Database 2021

- Measures financial inclusion
- Source: World Bank
- Columns = 1232 indicators of inclusion (% of population owning a bank account, debit card, taking loans, etc)
- Rows are countries
- Contains 4 years of data for each country (2011, 2014, 2017, 2021)

Country code	Year	Adult population	Region	Income group	Account (% age 15+)
AFG	2011	15124473	South Asia	Low income	9%
AFG	2014	17300802	South Asia	Low income	10%
AFG	2017	19718214	South Asia	Low income	15%
AFG	2021	22647496	South Asia	Low income	10%
ALB	2011	2258900	Europe & Central Asia (excluding high income)	Upper middle income	28%
ALB	2014	2320235	Europe & Central Asia (excluding high income)	Upper middle income	38%
ALB	2017	2347367	Europe & Central Asia (excluding high income)	Upper middle income	40%
ALB	2021	2348634	Europe & Central Asia (excluding high income)	Upper middle income	44%
DZA	2011	26161802	Middle East & North Africa (excluding high income)	Lower middle income	33%
DZA	2014	27492610	Middle East & North Africa (excluding high income)	Lower middle income	50%
DZA	2017	28736970	Middle East & North Africa (excluding high income)	Lower middle income	43%
DZA	2021	30352150	Middle East & North Africa (excluding high income)	Lower middle income	44%
AGO	2011	12369092	Sub-Saharan Africa (excluding high income)	Lower middle income	39%
AGO	2014	13734924	Sub-Saharan Africa (excluding high income)	Lower middle income	29%
ARG	2011	30168788	Latin America & Caribbean (excluding high income)	Upper middle income	33%
ARG	2014	31436980	Latin America & Caribbean (excluding high income)	Upper middle income	50%
ARG	2017	32658140	Latin America & Caribbean (excluding high income)	Upper middle income	49%
ARG	2021	34288808	Latin America & Caribbean (excluding high income)	Upper middle income	72%
ARM	2011	2317369	Europe & Central Asia (excluding high income)	Upper middle income	17%
ARM	2014	2327779	Europe & Central Asia (excluding high income)	Upper middle income	18%
ARM	2017	2341277	Europe & Central Asia (excluding high income)	Upper middle income	48%
ARM	2021	2345923	Europe & Central Asia (excluding high income)	Upper middle income	55%
AUS	2011	17838594	High income	High income	99%
AUS	2014	18752160	High income	High income	99%
AUS	2017	19606172	High income	High income	100%
AUS	2021	20731388	High income	High income	99%
AUT	2011	7132590	High income	High income	97%
AUT	2014	7271094	High income	High income	97%
AUT	2017	7498625	High income	High income	98%
AUT	2021	7632042	High income	High income	100%
AZE	2011	6986745	Europe & Central Asia (excluding high income)	Upper middle income	15%
AZE	2014	7286026	Europe & Central Asia (excluding high income)	Upper middle income	29%

% of Population with a Bank Account in 2021



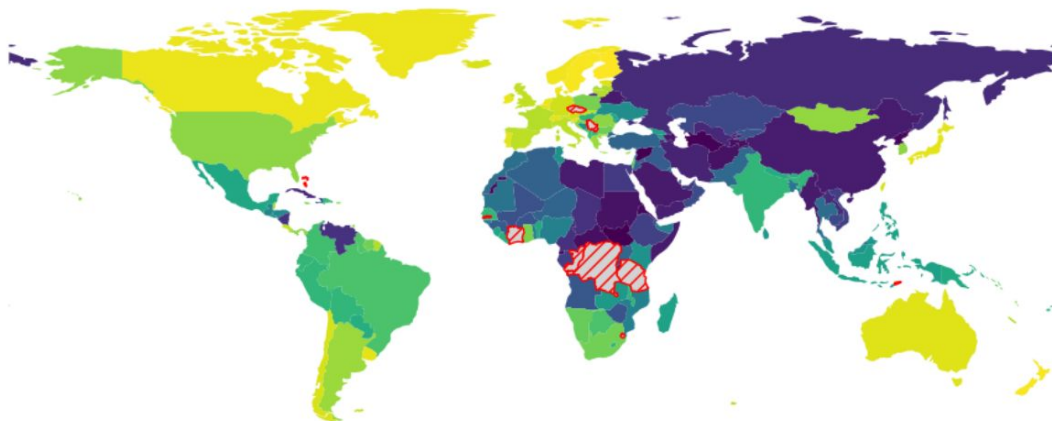
Data Sets

Freedom House Index

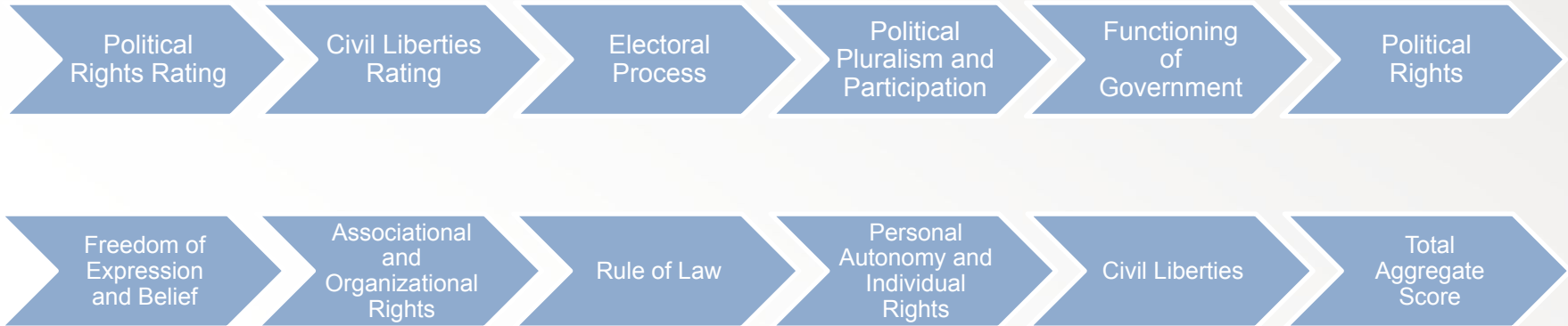
- Source: Freedom House
- 12 columns of freedom indicators (Political rights, civil liberties, political pluralism, etc)
- Rows are countries
- Data from 2006 to 2024 yearly basis
- ***Bias? Western Values***

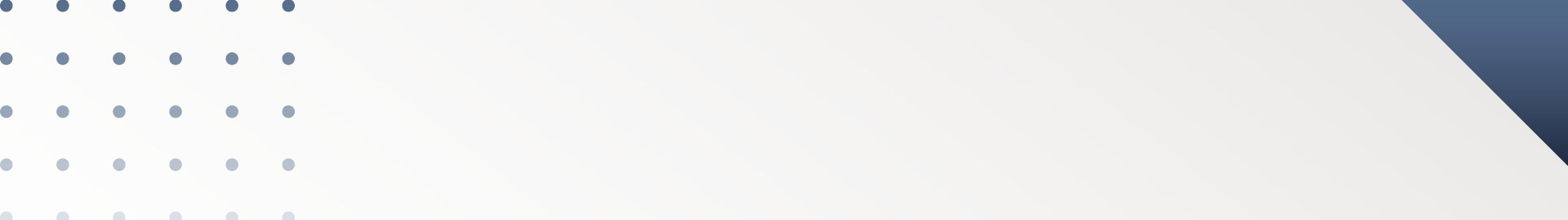
Country/Territory	Region	C/T?	Edition	Status	PR Rating	CL Rating	A	B	C	Add Q	Add A	PR	D	E	F	G	CL	Total
Cambodia	Asia	c	2024	NF	7	5	0	2	2	0	N/A	4	7	3	3	6	19	23
Cameroon	Africa	c	2024	NF	6	6	1	3	2	0	N/A	6	4	2	0	3	9	15
Canada	Americas	c	2024	F	1	1	12	16	11	0	N/A	39	15	12	15	16	58	97
Central African Rep	Africa	c	2024	NF	7	7	1	1	0	1	N/A	1	3	1	0	0	4	5
Chad	Africa	c	2024	NF	7	6	0	1	0	0	N/A	1	6	4	1	3	14	15
Chile	Americas	c	2024	F	1	1	12	15	11	0	N/A	38	16	12	13	15	56	94
China	Asia	c	2024	NF	7	6	0	0	1	3	N/A	-2	1	2	2	6	11	9
Colombia	Americas	c	2024	F	2	3	10	13	8	0	N/A	31	13	6	9	11	39	70
Comoros	Africa	c	2024	PF	5	4	5	7	4	0	N/A	16	8	6	6	6	26	42
Congo (Brazzaville)	Africa	c	2024	NF	7	6	0	2	0	0	N/A	2	6	2	1	6	15	17
Congo (Kinshasa)	Africa	c	2024	NF	7	6	1	2	1	0	N/A	4	7	3	1	4	15	19
Costa Rica	Americas	c	2024	F	1	1	12	15	11	0	N/A	38	16	11	13	13	53	91
Cote d'Ivoire	Africa	c	2024	PF	4	4	6	9	4	0	N/A	19	10	7	6	7	30	49
Crimea	Eurasia	t	2024	NF	7	7	0	0	0	2	N/A	-2	0	1	0	3	4	2
Croatia	Europe	c	2024	F	2	2	12	14	8	0	N/A	34	13	12	11	13	49	83
Cuba	Americas	c	2024	NF	7	6	0	0	1	0	N/A	1	4	0	1	6	11	12
Cyprus	Europe	c	2024	F	1	1	12	15	11	0	N/A	38	15	11	15	13	54	92
Czech Republic	Europe	c	2024	F	1	1	12	15	9	0	N/A	36	16	12	15	15	58	94
Denmark	Europe	c	2024	F	1	1	12	16	12	0	N/A	40	16	12	14	15	57	97
Djibouti	Africa	c	2024	NF	7	5	2	3	0	0	N/A	5	6	3	4	6	19	24
Dominica	Americas	c	2024	F	1	1	11	16	10	0	N/A	37	15	11	15	15	56	93
Dominican Republic	Americas	c	2024	PF	3	3	9	11	7	0	N/A	27	14	10	8	9	41	68
Eastern Donbas	Eurasia	t	2024	NF	7	7	0	0	0	1	N/A	-1	1	0	0	3	4	3
Ecuador	Americas	c	2024	PF	3	3	11	12	6	0	N/A	29	12	9	7	10	38	67
Egypt	Africa	c	2024	NF	6	6	2	2	2	0	N/A	6	3	1	2	6	12	18
El Salvador	Americas	c	2024	PF	4	4	9	10	2	0	N/A	21	11	7	4	10	32	53
Equatorial Guinea	Africa	c	2024	NF	7	7	0	0	0	0	N/A	0	2	0	0	3	5	5
Eritrea	Africa	c	2024	NF	7	7	0	0	1	0	N/A	1	1	0	0	1	2	3

Total Freedom in 2024



Freedom House Index: 12 Indicators

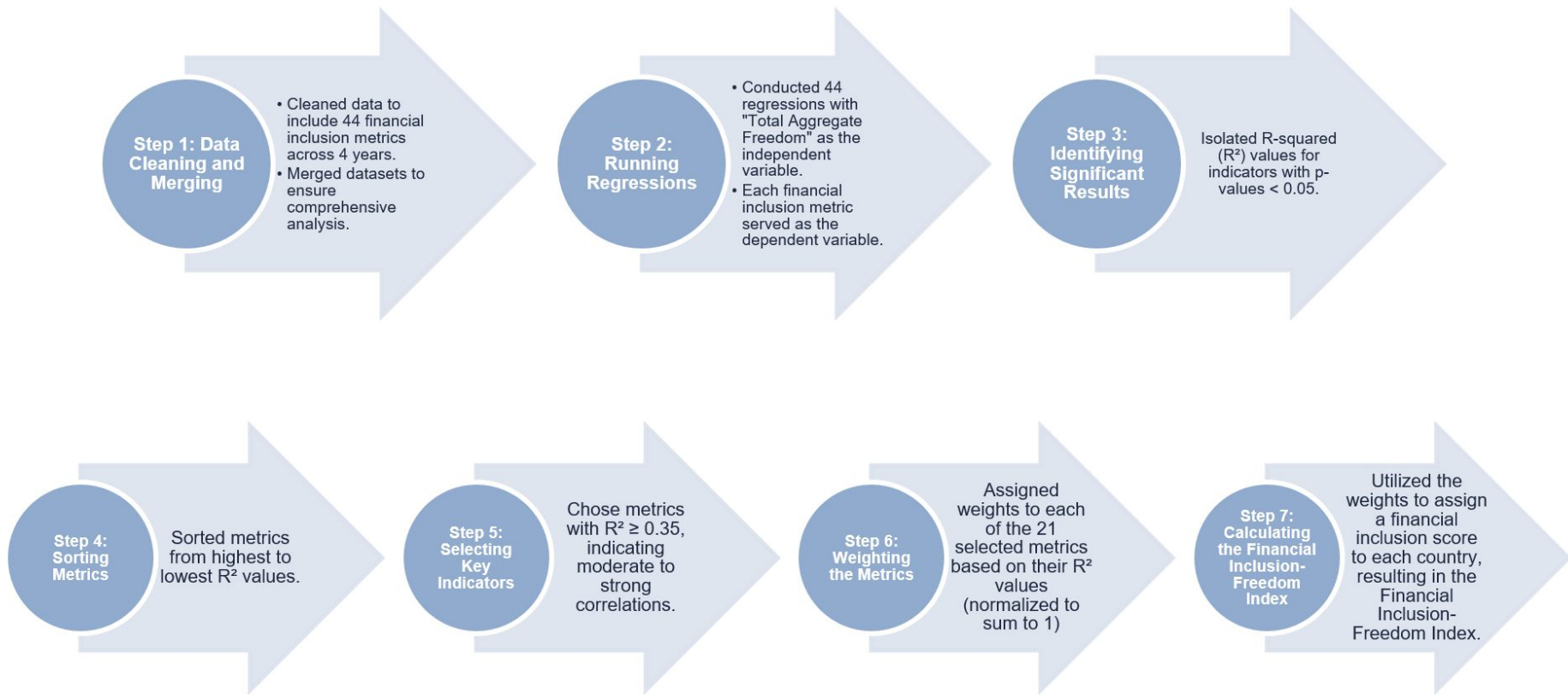




METHODOLOGY & ANALYSIS

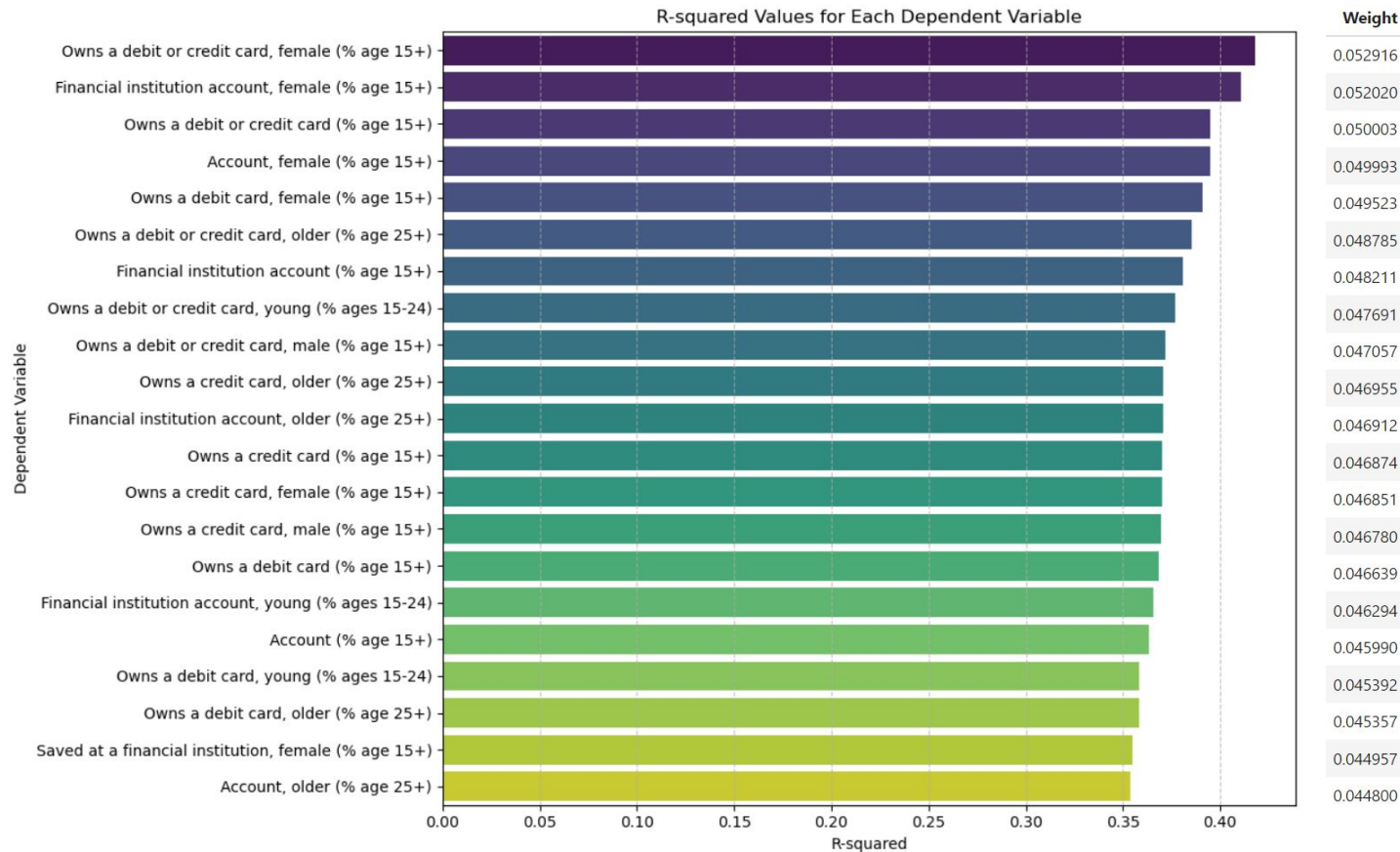
Cleaned Data Formatting

		Region_X	Country or Territory?	Freedom Status	Political Rights Rating	Civil Liberties Rating	Electoral Process	Political Pluralism and Participation	Functioning of Government	Political Rights category	Freedom of Expression and Belief	...	Saved at a financial institution, older (% age 25+)	Borrowed from a formal financial institution, female (% age 15+)	Borrowed from a formal financial institution, male (% age 15+)	Borrowed from a formal financial institution, young (% ages 15-24)	Borrowed from a formal financial institution, older (% age 25+)	Borrowed from family or friends, female (% age 15+)
Location Name	Edition																	
Afghanistan	2011	Asia	c	Not Free	6	6	3	6	1	10	5	...	0.03	0.05	0.1	0.05	0.09	0.21
	2014	Asia	c	Not Free	6	6	3	6	2	11	6	...	0.03	0.01	0.08	0.04	0.05	0.19
	2017	Asia	c	Not Free	6	6	2	6	2	10	6	...	0.04	0.03	0.04	0.01	0.05	0.24
	2021	Asia	c	Not Free	5	6	3	7	3	13	6	...	0.02	0.01	0.03	0.02	0.02	0.64
Albania	2011	Europe	c	Partly Free	3	3	8	11	7	26	13	...	0.09	0.06	0.09	0.05	0.08	0.13
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Zambia	2021	Africa	c	Partly Free	4	4	6	10	4	20	9	...	0.1	0.06	0.08	0.04	0.08	0.38
Zimbabwe	2011	Africa	c	Not Free	6	6	2	4	2	8	5	...	0.23	0.04	0.06	0.01	0.07	0.56
	2014	Africa	c	Not Free	5	6	3	6	3	12	6	...	0.06	0.04	0.04	0.02	0.06	0.55
	2017	Africa	c	Partly Free	5	5	3	6	3	12	8	...	0.06	0.04	0.06	0.03	0.05	0.37
	2021	Africa	c	Not Free	6	5	3	5	3	11	7	...	0.04	0.02	0.04	0.02	0.03	0.37

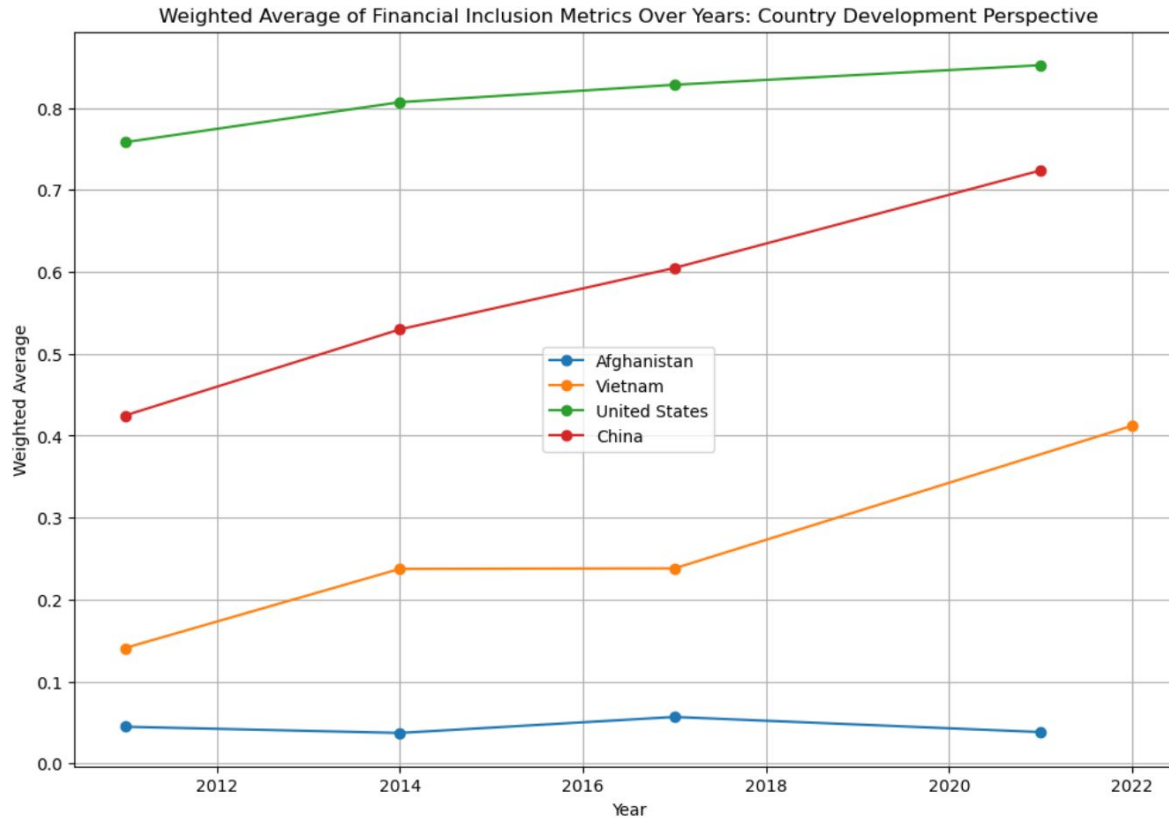


Part 1: Developing the Financial Inclusion-Freedom Index

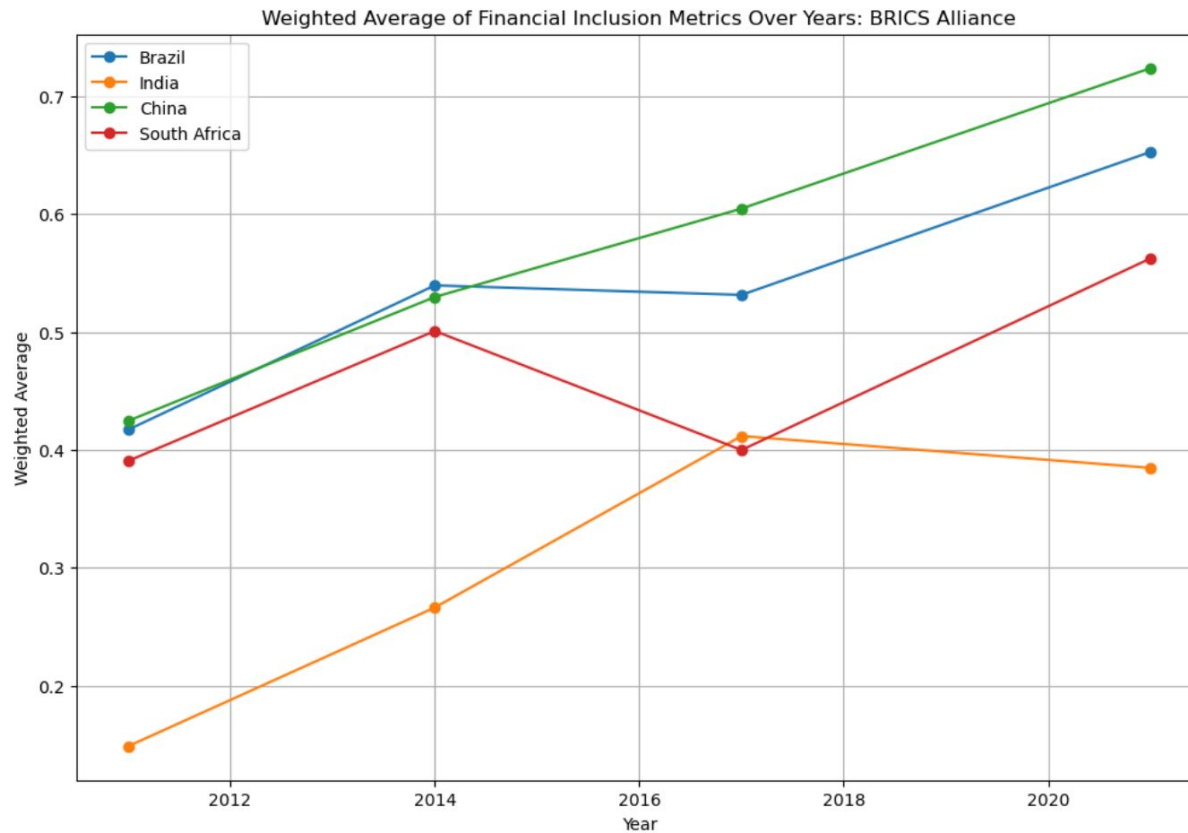
Cumulative definition of financial inclusion with respect to freedom to quantify how inclusive a country is



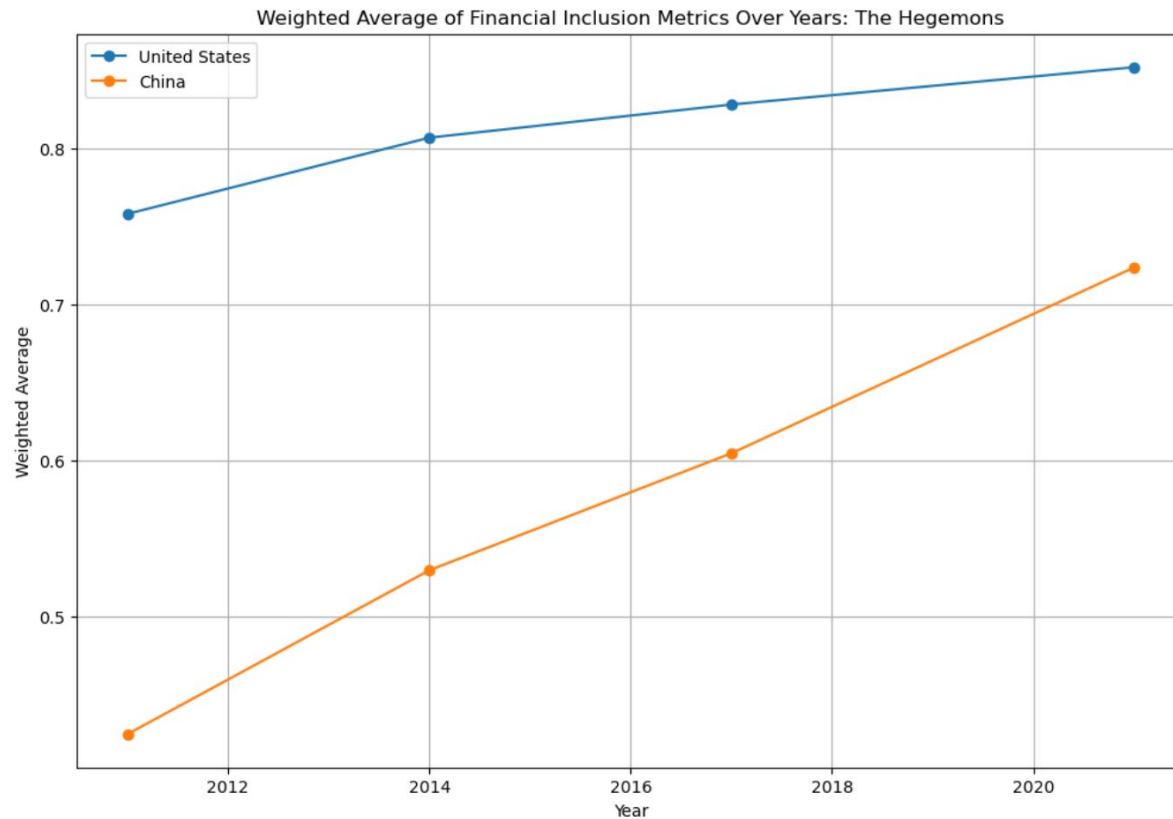
Part 1: Developing the Financial Inclusion–Freedom Index



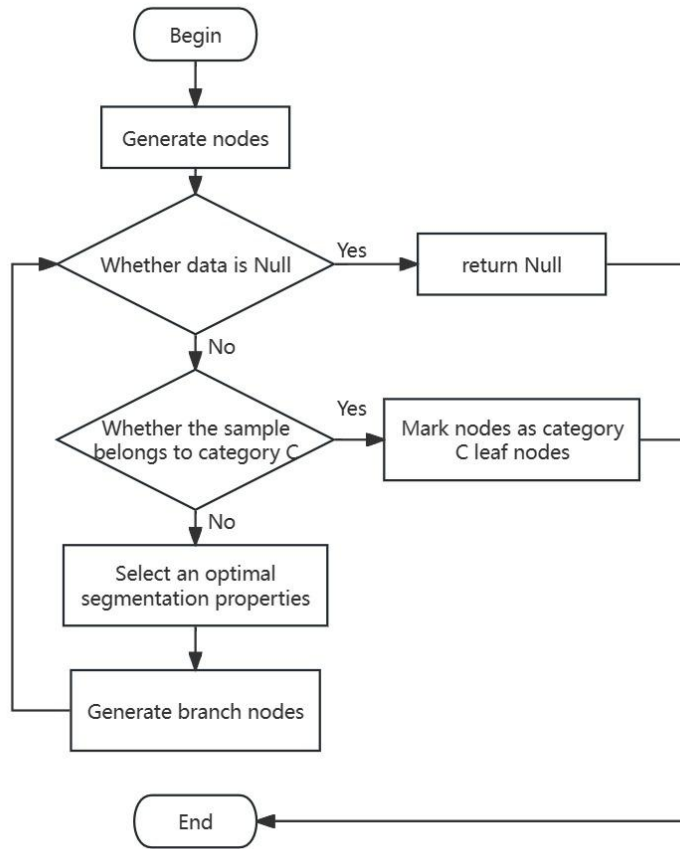
Part 1: Developing the Financial Inclusion-Freedom Index



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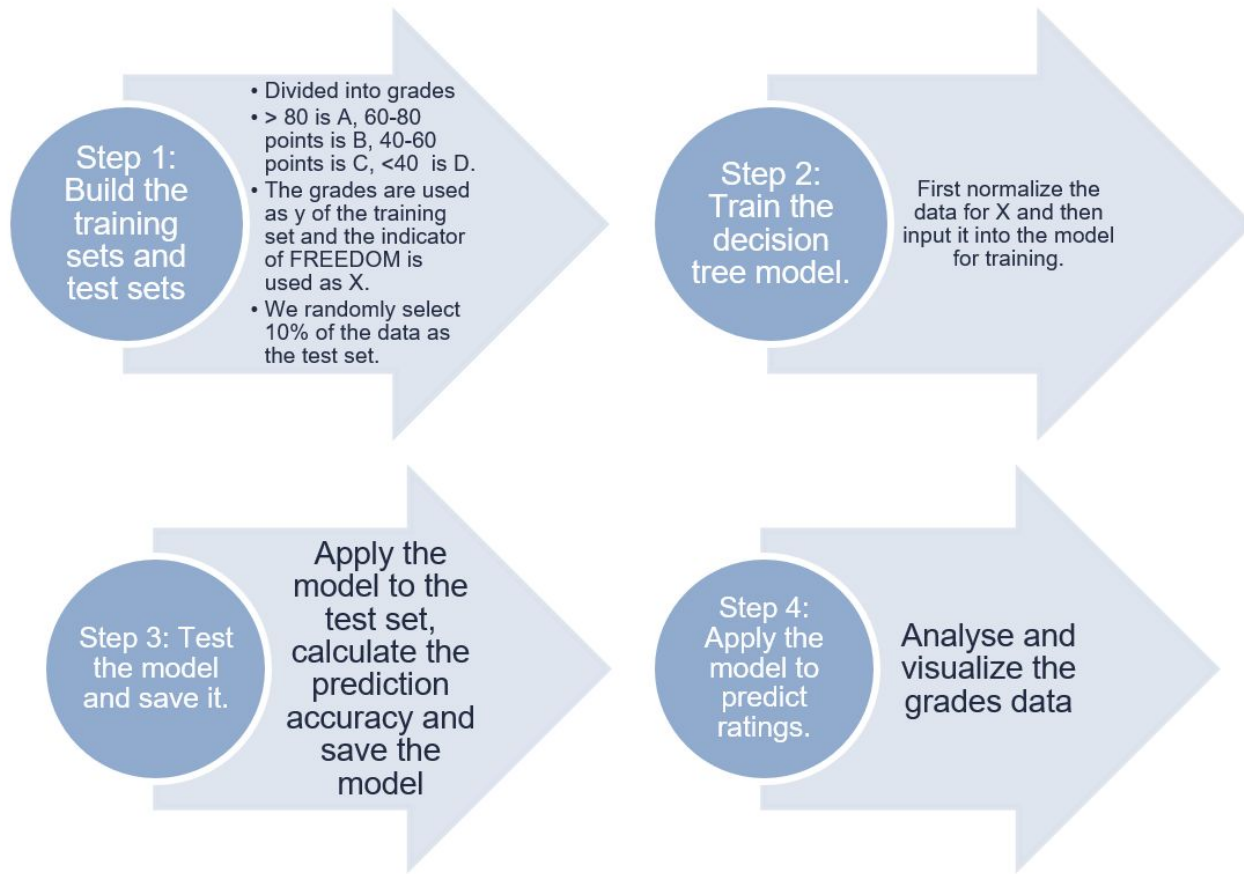
Part 1: Developing the Financial Inclusion-Freedom Index



The basic process of a decision tree

- **Feature Selection:** Selecting the best feature as a discriminator for the current node
- **Knot Splitting:** Split the current node into multiple child nodes based on selected features
- **Exit Condition:** When a certain stopping condition is met, the splitting stops and the current node is marked as a leaf node

Part 2: Machine Learning Model

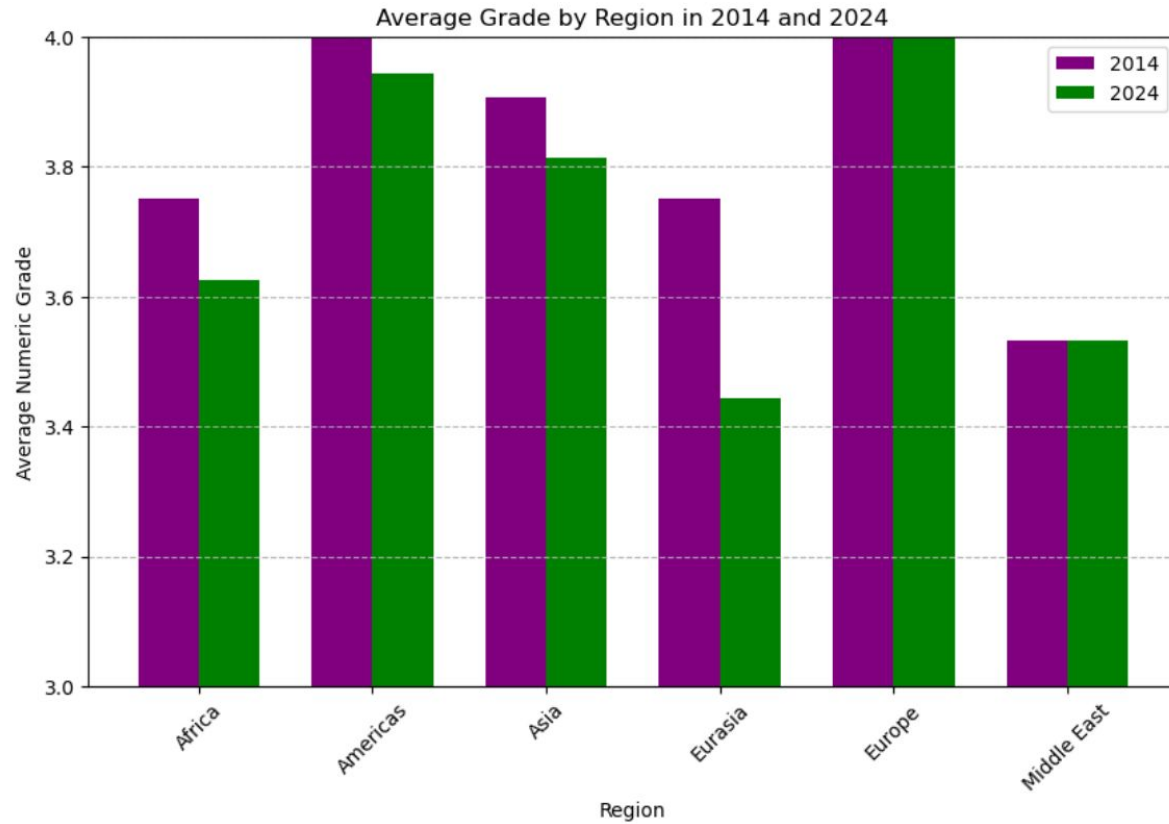


Part 2: Machine Learning Model

Prediction Results of Decision Tree

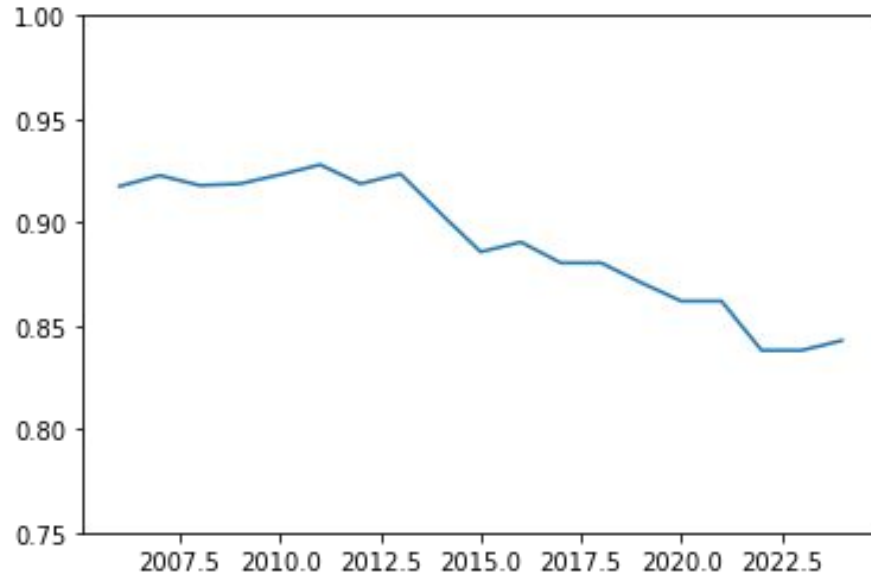
	Country/Territory	Region	Edition	grade
0	Abkhazia	Eurasia	2024	A
1	Afghanistan	Asia	2024	B
2	Albania	Europe	2024	A
3	Algeria	Africa	2024	A
4	Andorra	Europe	2024	A
...	...	...	...	...
3964	Vietnam	Asia	2006	A
3965	Western Sahara	Africa	2006	B
3966	Yemen	Middle East	2006	A
3967	Zambia	Africa	2006	A
3968	Zimbabwe	Africa	2006	C

3969 rows × 4 columns



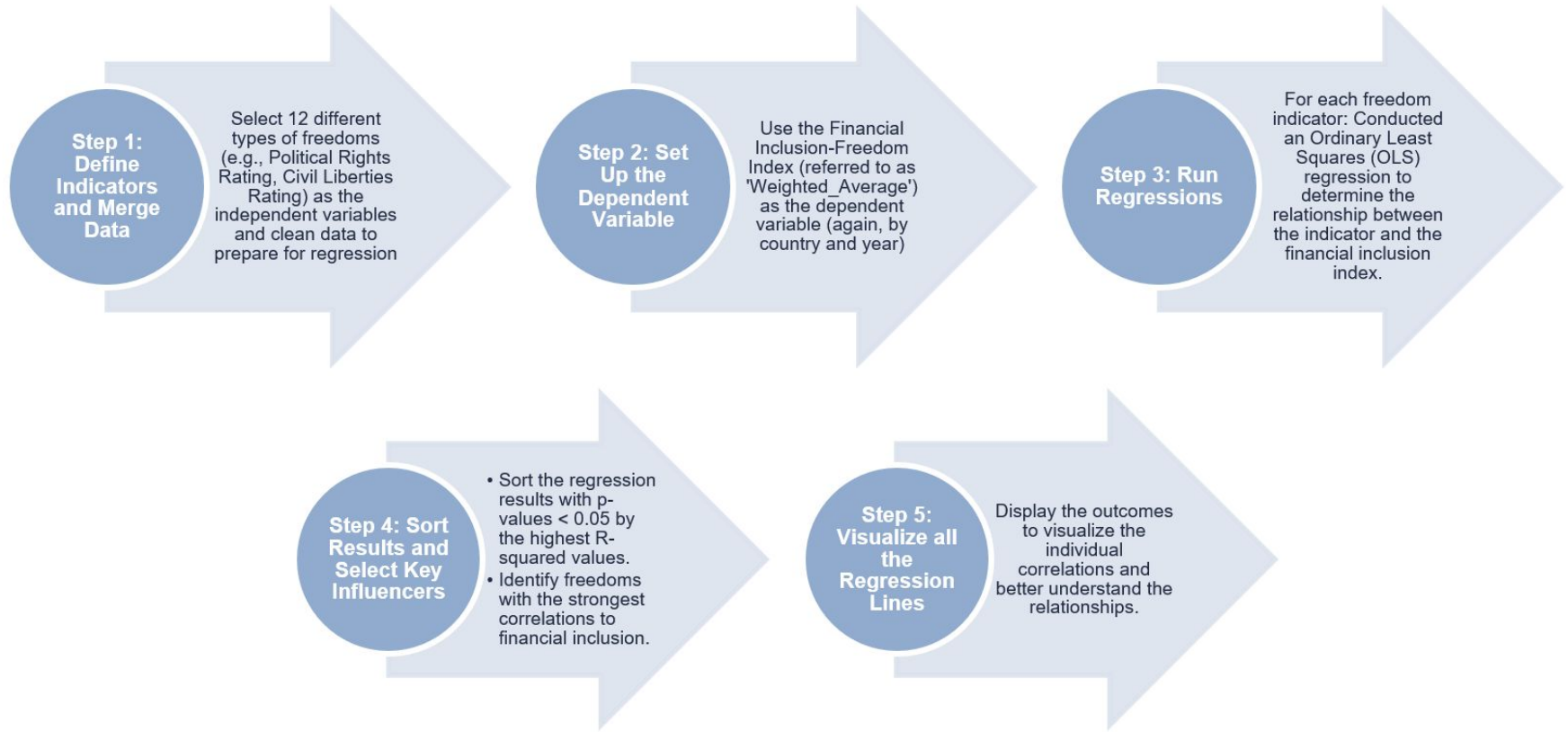
Part 2: Machine Learning Model

Why a downward trend over time?

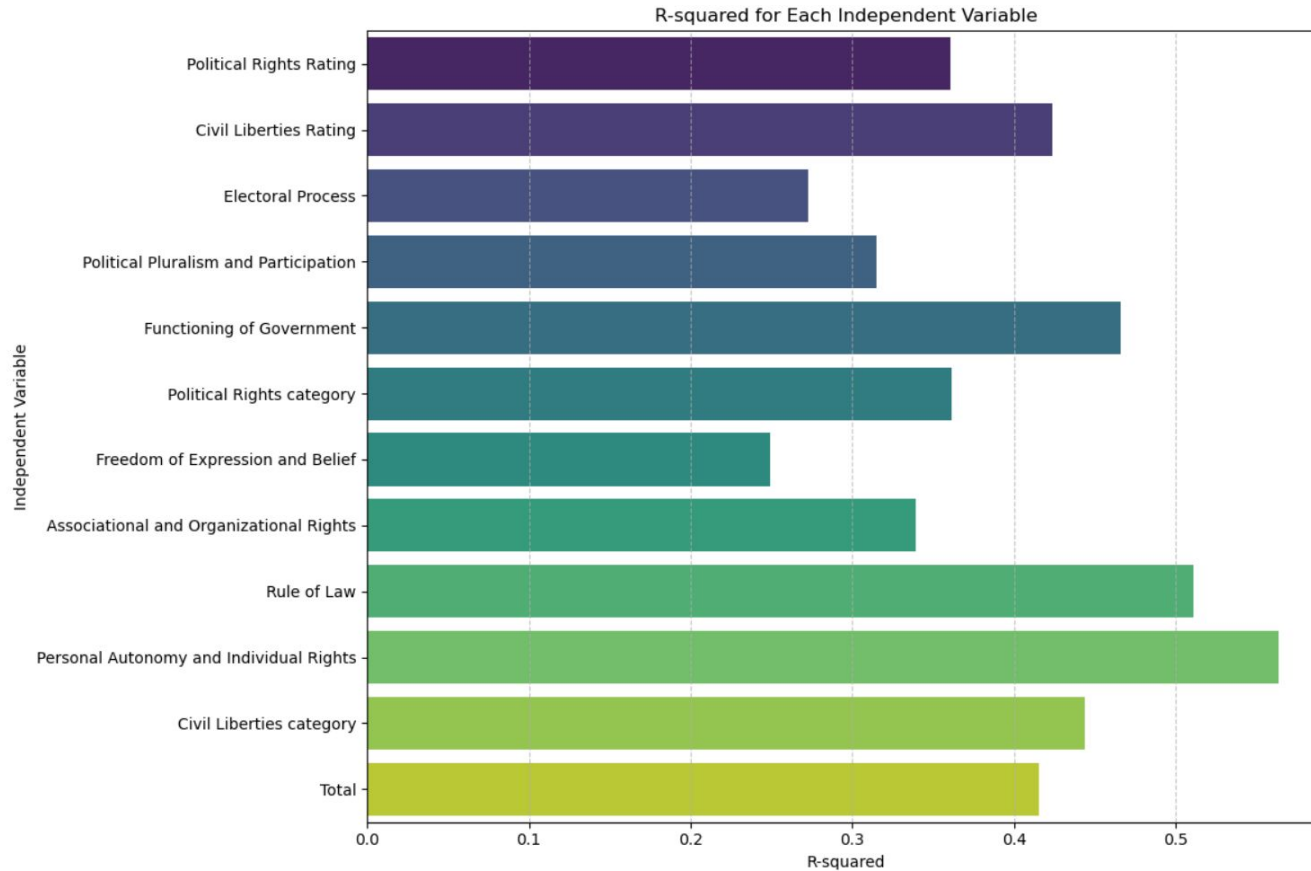


- **Widening of the gap between the rich and the poor that accompanies economic growth**
- **Reduced or unequal employment opportunities**
- **Impact of globalisation**
-

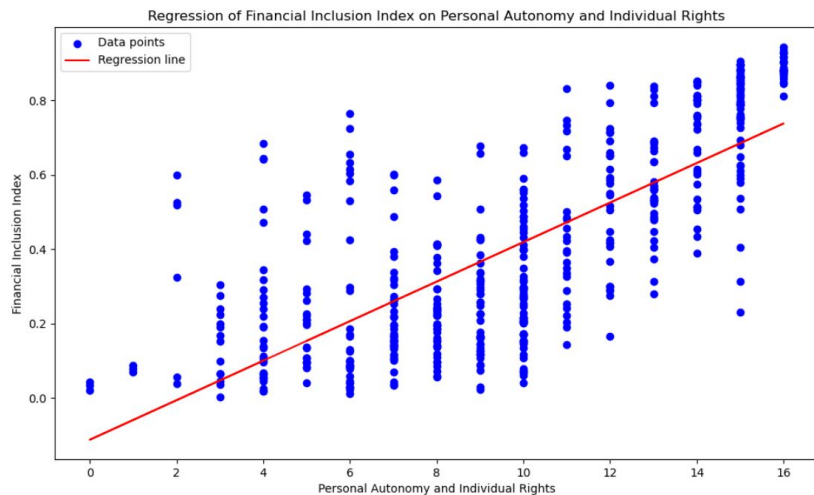
Part 2: Machine Learning Model



Part 3: Identifying Role of Different Freedoms on Inclusion



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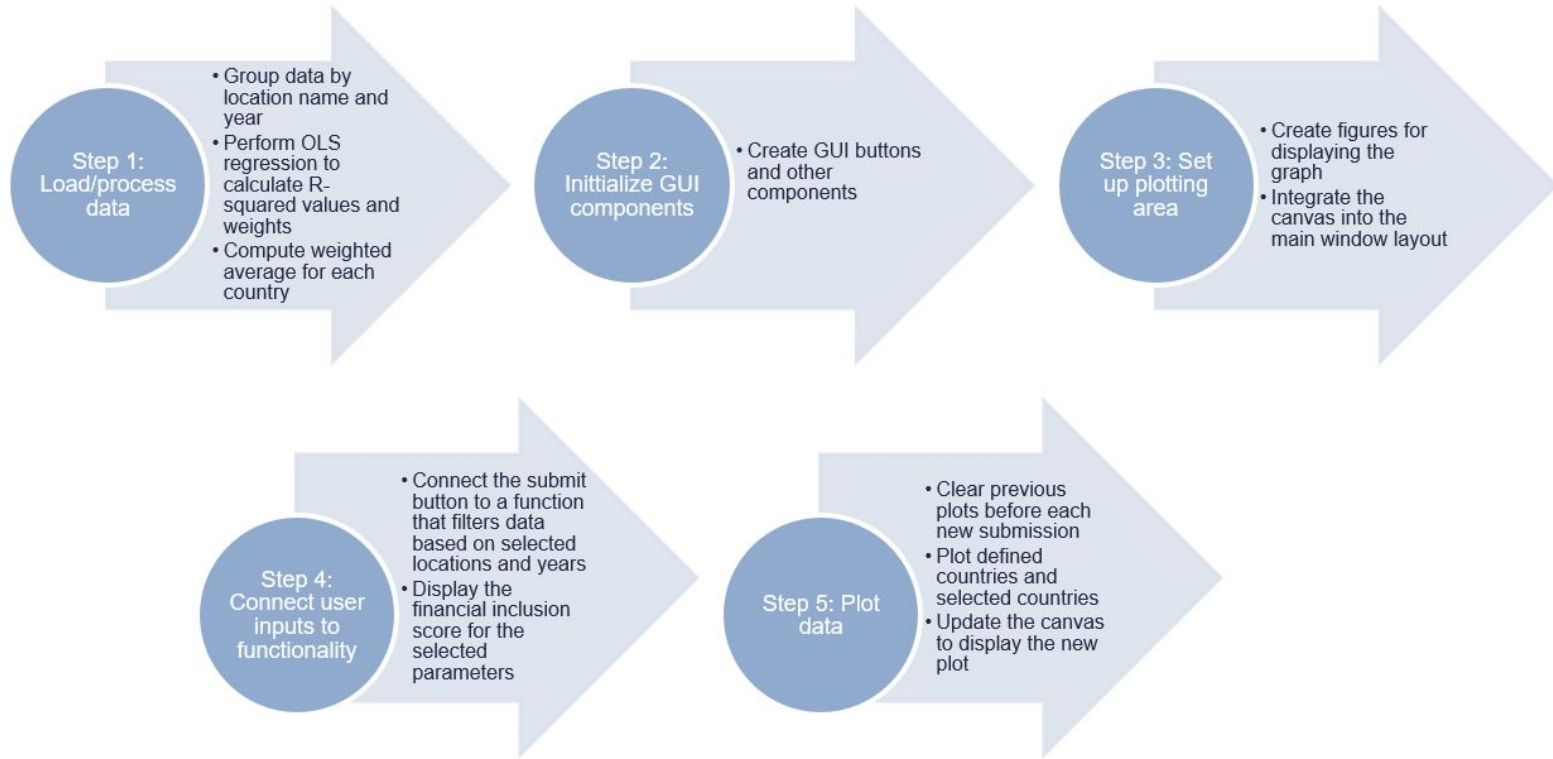


Smallest $R^2 \rightarrow$

$\leftarrow$ Largest R^2



Part 3: Identifying Role of Different Freedoms on Inclusion



Part 4: Displaying Our Results and Creating a GUI Interface

```

# Predefined countries to plot
countries = ['Afghanistan', 'United States', 'China']

def plot_selected_country(selected_country, ax):
    if selected_country not in countries:
        country_data = merged_df.xs(selected_country)
        ax.plot(country_data.index, country_data['Weighted_Average'], marker='o', label=selected_country)

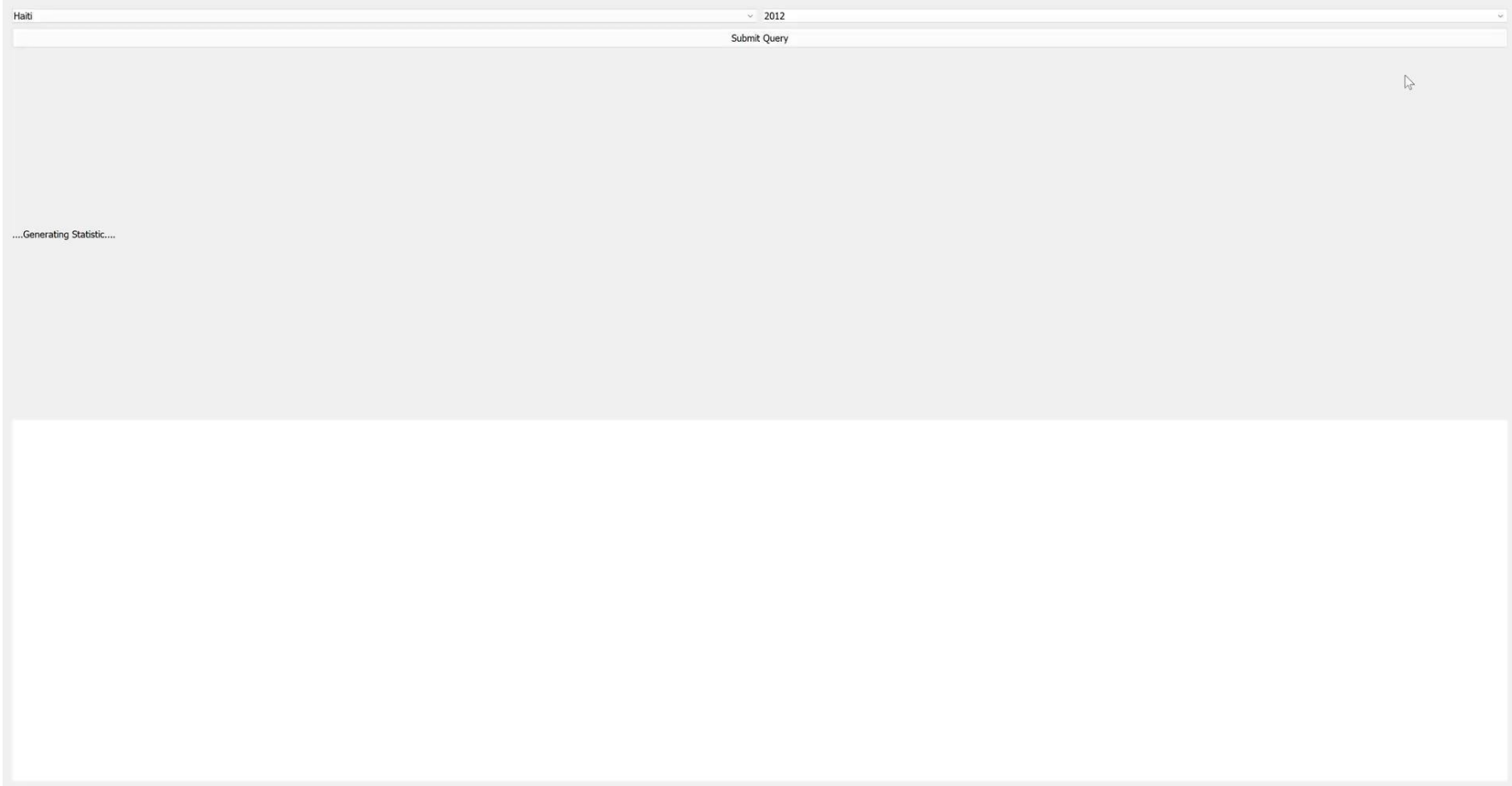
class MainWindow(QWidget):
    def __init__(self):
        super().__init__()
        self.setWindowTitle('Freedom Index impacts on Financial Inclusion')
        file_path = r"C:\Users\maya\OneDrive\Desktop\gatech\Study abroad\Machine Learning"
        self.data = pd.read_excel(file_path, sheet_name='Full Data Set')
        self.data['Year'] = self.data['Year'].astype(str)
        self.outer = QVBoxLayout()
        self.inner1 = QHBoxLayout()
        self.location = QComboBox()
        self.location.addItem('United States')
        self.location.addItem('Afghanistan')
        self.location.addItem('China')
        self.year = QComboBox()
        self.year.addItem('2000')
        self.year.addItem('2001')
        self.year.addItem('2002')
        self.year.addItem('2003')
        self.year.addItem('2004')
        self.year.addItem('2005')
        self.year.addItem('2006')
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        self.year.addItem('2024')
        self.year.addItem('2025')
        self.year.addItem('2026')
        self.year.addItem('2027')
        self.year.addItem('2028')
        self.year.addItem('2029')
        self.year.addItem('2030')
        self.inner1.addWidget(self.location)
        self.inner1.addWidget(self.year)
        self.outer.addLayout(self.inner1)
        self.query = QPushButton("Submit Query")
        self.outer.addWidget(self.query)
        self.output = QLabel('...Generating Statistic...')
        self.outer.addWidget(self.output)
        # Create a Figure instance
        self.fig = Figure()
        # Create a FigureCanvas instance
        self.canvas = FigureCanvas(self.fig)
        self.outer.addWidget(self.canvas)
        self.setLayout(self.outer)
        self.query.clicked.connect(self.filterframe)

    def filterframe(self):
        data = self.data
        location = self.location.currentText()
        year = self.year.currentText()
        # Filter data based on selected inputs
        reduced_data = data[(data['Location Name'] == location) & (data['Year'] == year)]
        if reduced_data.empty:
            self.output.setText(f'No data available for {location} in the year {year}.')
            return
        financial_inclusion_score = reduced_data['Financial Inclusion score'].values[0]
        self.output.setText(f'The Financial Inclusion Score for ' + f"{location}" + " in the year " + f"{year}" + ' is ' + f"{financial_inclusion_score}")
        # Clear previous plot
        self.fig.clear()
        # Create a new plot
        ax = self.fig.add_subplot(111)
        # Plot predefined countries
        for country in countries:
            country_data = merged_df.xs(country)
            ax.plot(country_data.index, country_data['Weighted_Average'], marker='o', label=country)
        # Plot selected country
        plot_selected_country(location, ax)
        ax.set_xlabel('Year')
        ax.set_ylabel('Weighted Average')
        ax.set_title('Total of Financial Inclusion Metrics Over Years')
        ax.legend()
        ax.grid(True)
        # Draw the plot
        self.canvas.draw()

# Create an instance of QApplication if it does not already exist
app = QApplication.instance()
if not app:
    app = QApplication(sys.argv)
main = MainWindow()
main.show()
# Ensure the event loop is started if not already running
if not app.exec_():
    sys.exit()

```

Part 4: GUI Code



Part 4: Displaying Our Results and Creating a GUI Interface

Conclusion

1

Key Insights: Strong positive correlation between political freedoms and financial inclusion metrics across various countries

2

Impacts of Freedoms: Specific political freedoms, such as civil liberties and rule of law, have been identified as strong predictors of financial inclusion

3

Machine Learning Success: The decision model effectively categorized each country's financial inclusivity score by year

4

Future applications: The developed GUI tool can aid policymakers and researchers in visualizing and understanding the relationship between political freedoms and financial inclusion for better decision making

THANKS!



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